Date	03 July 2026
Team ID	SWTID-2026-4799
Team Size	5
Team Leader	Hari Bellamkonda
Team Member 1	Challa Sukumar
Team Member 2	Umesh CHALAPADHI
Team Member 3	Girish Kumar
Team Member 4	Vignesh A
Project Name	Machine Learning—Based Credit Card Approval Prediction System
Maximum Marks	5 Marks

S.NO	Code Quality Parameter	Description	Followed (Yes / NO / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	PEP 8 standards followed
2	Proper File Structure	Files and folders are logically organized	Yes	Clear separation of modules, templates, static and models
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names
4	Function / Method Names	Functions are descriptively named	Yes	Functions named as per their responsibility
5	Code Comments	Inline and block comments explain logic	Yes	Important logic is documented
6	Modular Design	Code is split into reusable functions/ modules	Yes	Reusable functions and modular scripts used

7	NO Redundant Code	Duplicate or unused code is removed	Yes	No redundant code found

S. NO	Component I Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module (preprocess.py)	Python, Pandas, Scikitlearn	Used in train_model.py and predict.py	High
2	Prediction Module (predict.py)	Python, Scikit-learn	Used for model inference and prediction generation	High
3	Database Module	Python, MySQL	Used for user authentication, application records, and prediction history	High
4	Frontend Templates	HTML5, CSS3, Bootstrap 5, JavaScript	Used across Login, Registration, Dashboard, and Prediction pages	High
5	Machine Learning Model	Scikit-learn, Joblib	Used by the prediction engine to classify applicant approval status	High

Code-Layout, Readability and Reusability

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

Readability	5	Meaningful variable names, consistent formatting, and readable coding practices following Python standards.

8	Error Handling	Exceptions and errors are handled gracefully	Yes	Try-except blocks used wherever required

Reusable Components / Modules:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with clear separation of frontend, backend, database, and machine learning components.

Overall Code Quality Assessment:

Reusability	5	Reusable modules for preprocessing, prediction, authentication, and database operations improve maintainability.
Documentation / Comments	5	Source code contains appropriate comments and documentation to improve understanding and future maintenance.

Overall Score	5/5	The project demonstrates excellent code organization, readability, modularity, maintainability, and reusability following Software Engineering best practices.